Assignment:3

Problem Statement: -

Design and develop inheritance for a given case study, identify objects and relationships and implement inheritance wherever applicable. Employee class with Emp_name, Emp_id, Address, Mail_id, and Mobile_no as members. Inherit the classes, Programmer, Team Lead, Assistant Project Manager and Project Manager from employee class. Add Basic Pay (BP) as the member of all the inherited classes with 97% of BP as DA, 10 % of BP as HRA, 12% of BP as PF, 0.1% of BP for staff club fund. Generate pay slips for the employees with their gross and net salary.

Objectives:

1) To Study Inheritance and its types
2) To implement inheritance using OOP language

Theory:-

Inheritance:

Inheritance is a mechanism in which one object acquires all the properties and behaviors of a parent object. It is an important part of OOPs (Object Oriented programming system). The idea behind inheritance in Java is that you can create new classes that are built upon existing classes. When you inherit from an existing class, you can reuse methods and fields of the parent class. Moreover, you can add new methods and fields in your current class also.

Inheritance represents the **IS-A relationship** which is also known as a *parent-child* relationship.

The Need Inheritance:

- **Code Reusability:** The code written in the Superclass is common to all subclasses. Child classes can directly use the parent class code.
- **Method Overriding:** Method Overriding is achievable only through Inheritance. It is one of the ways by which Java achieves Run Time Polymorphism.
- **Abstraction:** The concept of abstract where we do not have to provide all details is achieved through inheritance. Abstraction only shows the functionality to the user.

Terms used in Inheritance

- **Class:** A class is a group of objects which have common properties. It is a template or blueprint from which objects are created.
- **Sub Class/Child Class:** Subclass is a class which inherits the other class. It is also called a derived class, extended class, or child class.
- **Super Class/Parent Class:** Superclass is the class from where a subclass inherits the features. It is also called a base class or a parent class.

- **Reusability:** As the name specifies, reusability is a mechanism which facilitates you to reuse the fields and methods of the existing class when you create a new class. You can use the same fields and methods already defined in the previous class.

Synatx:

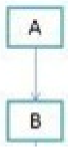
The **extends keyword** is used for inheritance in Java. Using the extends keyword indicates you are derived from an existing class. In other words, “extends” refers to increased functionality.

Syntax :

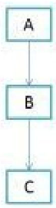
```
class derived-class extends base-class
{
   //methods and fields
}
```

Types of inheritance in java

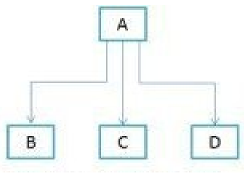
❖ **Single Inheritance:** When a class extends another one class only then we call it a single inheritance. The below flow diagram shows that class B extends only one class which is A. Here A is a parent class of B and B would be a child class of A.



❖ Multilevel Inheritance: Multilevel inheritance refers to a mechanism in OOP technology where one can inherit from a derived class, thereby making this derived class the base class for the new class. As you can see in below flow diagram C is subclass or child class of B and B is a child class of A.



❖ Hierarchical Inheritance: In such kind of inheritance one class is inherited by many sub classes. In below example class B,C and D inherits the same class A. A is parent class (or base class) of B,C & D.



Steps :

1. Start
2. Create the class Employee with name, Empid, address, mailid, mobileno as data members.
3. Inherit the classes Programmer, Team Lead, Assistant Project Manager and Project Manager from employee class.
4. Add Basic Pay (BP) as the member of all the inherited classes.
5. Calculate DA as 97% of BP, HRA as 10% of BP, PF as 12% of BP, Staff club fund as 0.1% of BP.
6. Calculate gross salary and net salary.
7. Generate payslip for all categories of employees.
8. Create the objects for the inherited classes and invoke the necessary methods to display the Payslip
9. Stop

Input:
Empid, address, mailid, mobileno, Basic Pay (BP)

Output:
gross and net salary slip

Question

1. Justify inheritance/ class relationship with classes Surgeon and Doctor.
2. Explain why multiple inheritance is not supported by Java.
3. Compare Composition and Inheritance in OOP.
4. What are different types of Inheritance supported by Java?
5. Why multiple inheritance is not supported by Java?
6. How to use Inheritance in Java?
7. What is the difference between Composition and Inheritance in OOP?
8. What is Super Keyword in Java?
   (write answer for the above questions) & Attached code with output